Q1)

1)

$$H(z) = \frac{(1+0{,}2z^{-1})(1-9z^{-1})}{1+0{,}81z^{-2}}$$

$$H(z) = \frac{(z+0{,}2)(z-9)}{z^2+0{,}81}$$

$z^2 = -0{,}81$

$z = \pm j\sqrt{0{,}81} = \pm j0{,}9$

$|j0{,}9| = 0{,}9 < 1$

Both poles are inside the unit circle

final answer:

Yes, system is stab

2)

zeros:

$1+0{,}2z^{-1} \Rightarrow z = -0{,}2$

$1-9z^{-1} \Rightarrow z = 9$

$z = 0{,}2 \rightarrow$ inside ✓

$z = 9 \rightarrow$ outside ✗

one min phase zero at $z = -0{,}2$

one non-min phase zero at $z = 9$

we need reflect zero at 9 inside unit circle to form $H_1(z)$,

Reflected zere of $z = 9$

$$z = \frac{1}{9}$$

So:

$$H_1(z) = \frac{(1+0{,}2z^{-1})(1-\frac{1}{9}z^{-1})}{1+0{,}81z^{-2}}$$

$$H_{ap}(z) = \frac{1-9z^{-1}}{1-\frac{1}{9}z^{-1}}$$

This is all pass filter because

- zeros and poles are reciprocal
- Magnitude response is 1 at all frequencies.

Min phese $H_1(z) = [(1+0{,}2z^{-1})] \cdot (1-((1/9)z^{-1}) / [1+0{,}81z^{-2})$

All pass system $H\_ap(z) = (1-9z^{-1}) / (1-(1/9)z^{-1})$

Therefore:

$$H(z) = H_1(z) \cdot H\_ap(z)$$

Q2)

1) Real valued impulse response

Answer:

If a system has a real-valued impulse response $h[n]$, then its poles and zeros must occur in conjugate pairs in z-plane. That is, if z is a pole or zero, then z^* must also be present.

2) finite impulse response (FIR)

FIR systems have no poles except possibly at the origin. In the z plane, all poles are at $z=0$, or there are no poles at all. Zeros can be anywhere.

3) $h[n] = h[2\alpha - n]$ where 2α is an integer

This indicates symmetry about $n = \alpha$, meaning the impulse response is even-symmetric around a midpoint.

In the z-plane, this symmetry implies that for every zero at z_0, there is a zero at $1/z_0$

4) Minimum phase

A minimum-phase system has all its zeros inside the unit circle in z plane.

5) All-pass

An all-pass system has zeros and poles that are reciprocal conjugates: for every zero at z_0, there is a pole at $1/z_0^*$. This causes the system to have constant magnitude. $(|H(e^{jw})| = 1)$, but variable phase.